

Notes on Scaling the Quasi-Symmetry DMRG Workflow to Larger Orbital Spaces

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1 Purpose

The current workflow is practical for the full STO-3G calculations considered so far, whose largest examples contain ten spatial orbitals and twenty qubits. It already avoids several unnecessary costs: DMRG is performed entirely with `pyblock2`, CISD determinants are added to the initial MPS in bounded batches, Clifford circuits and qubit permutations act directly on the MPS, and each Hamiltonian representation is processed and saved separately.

Larger basis sets introduce qualitatively different bottlenecks. The main issue is not DMRG alone. Full-CI reference states, entropy calculations, CISD-to-MPS conversion, symmetry scoring, transformed Pauli Hamiltonians, and MPO construction can each become the limiting step. This note records a possible sequence of changes rather than prescribing a final implementation.

2 Remove full CI from the required workflow

The size of the full-CI space grows combinatorially with the numbers of electrons and orbitals. Full CI should therefore become an optional small-system validation method rather than a required input.

For larger systems, a tightly converged spin-adapted fermionic DMRG calculation can provide both the reference energy and reference MPS. The reference energy used in the chemical-accuracy test would then be labeled by its source, for example FCI, selected CI, variational DMRG, or extrapolated DMRG. The reported criterion remains

$$|E(\chi) - E_{\text{ref}}| \leq 1.6 \text{ mHa}, \tag{1}$$

but it no longer assumes that E_{ref} is available from FCI.

The software interface should consequently accept a general reference-state object. For small systems this may be a sparse CI vector; for larger systems it should normally be a saved MPS together with its energy, quantum numbers, and convergence information.

3 Calculate entropies from an MPS

The current preprocessing calculates entropies and mutual information from a saved FCI state. Instead, one- and two-site reduced density matrices can be contracted directly from a converged reference MPS. This gives the one-site entropies and pairwise mutual information needed for Fiedler ordering without constructing a vector of length 2^{n_q} .

For a transformed representation, the same unitary circuit used for the Hamiltonian can be applied to a copy of the reference MPS. Entropies can then be measured from the transformed MPS. This procedure applies equally to Clifford circuits, qubit permutations, and orbital rotations decomposed into Givens rotations. Every entropy result should record the MPS bond dimension and discarded weight used in this preprocessing step.

4 Make the orbital space explicit

In a larger basis, retaining every orbital will often be neither necessary nor affordable. Molecular input and saved metadata should explicitly record the basis, frozen orbitals, active orbitals, active electron number, target spin, and original orbital ordering. A useful interface should support both the full orbital space and a selected active space without changing the subsequent symmetry and DMRG code.

The active-space definition must also be carried through the CISD state, fermionic MPO, Jordan–Wigner Hamiltonian, and reference energy. Silent differences between these spaces would invalidate energy comparisons.

5 Improve CISD-to-MPS conversion

Although CISD is polynomial in system size, its number of double excitations scales approximately as

$$O(n_{\text{occ}}^2 n_{\text{virt}}^2). \quad (2)$$

The present method retains every CISD determinant, sorts them by coefficient magnitude, adds them sequentially to an MPS, and compresses the growing sum in batches. This is reliable for the current systems but may become slow for large virtual spaces.

Possible improvements include balanced-tree addition of determinant MPSs, adaptive batch sizes, an intermediate bond-dimension cap, and construction by applying single- and double-excitation operators to the Hartree–Fock MPS. Disk-backed intermediate tensors may also be useful. Selected-CI or coefficient truncation can be offered as an explicit optional approximation, but the retained norm, discarded norm, and change in energy must be reported.

6 Avoid large intermediate Pauli operators

An electronic Hamiltonian contains $O(n_{\text{orb}}^4)$ integral terms. Jordan–Wigner mapping and Clifford conjugation can make the explicit Pauli representation expensive even when the corresponding fermionic MPO remains manageable. Materializing the complete transformed `QubitOperator` and all intermediate MPO sums may therefore become the dominant memory cost.

The transformation and MPO builder should eventually operate as a streaming pipeline: transform terms in chunks, combine identical Pauli words immediately, and release each chunk after it has been incorporated into the MPO. Coefficient screening should be optional and controlled by an explicit error tolerance. Density fitting, Cholesky decomposition, and other factorized representations of the two-electron interaction should also be investigated, especially if the factorized structure can be preserved during MPO construction.

For sufficiently large orbital spaces, a Pauli-space DMRG calculation may remain much more expensive than the spin-adapted fermionic calculation. That difference is scientifically relevant and should not be hidden by using different uncontrolled truncations in the two representations.

7 Scale the symmetry search

The present beam-search safeguards—50 pairwise seed terms and a candidate pool capped at 1000 operators—limit the combinatorial part of the search. The remaining cost comes from repeatedly scoring candidates against a large Hamiltonian and state.

Useful improvements include caching Pauli commutation relations in binary symplectic form, evaluating candidate scores incrementally, screening small Hamiltonian terms during pool generation, and parallelizing independent candidate evaluations. The initial candidate pool, its ranking scores, and the state of a partially completed search should be saved so that a long search can be resumed exactly.

When the Hamiltonian and state are transformed by a known unitary, objective values can sometimes be evaluated in the original frame by transforming only the candidate Pauli operator back to that frame. For Clifford transformations this maps a Pauli product to another Pauli product. It can avoid constructing a large sparse matrix for every transformed Hamiltonian and is especially useful for repeated symmetry searches.

8 Control the cost of applying unitary circuits

Applying a long circuit to an MPS can increase its bond dimension before DMRG begins. The circuit layer should report the discarded singular-value weight, the maximum intermediate bond dimension, and the energy error introduced by compression. Separate bond caps may be appropriate for reference-state transformation and for the later DMRG calculation.

Additional improvements include reordering commuting gates, reducing swap routing, applying independent gates in layers, and variationally compressing the state after the complete circuit. These operations should use the common unitary interface so that Clifford transformations and orbital rotations follow the same execution and validation path.

9 Exploit exact symmetries where available

The fermionic calculations already use particle-number and spin-adapted $SU(2)$ symmetry. The general Pauli calculations currently do not enforce particle number or total spin. For larger systems, exact mapped symmetries could be used to restrict the qubit calculation or eliminate fixed qubits. Approximate symmetries cannot be fixed without introducing an additional approximation, so exact and approximate generators must remain clearly distinguished.

10 Suggested implementation order

The following order removes the most restrictive bottlenecks first:

1. introduce a reference-state interface supporting saved MPSs;
2. compute entropies and mutual information directly from an MPS;
3. generalize molecular input and active-space metadata;
4. improve and instrument CISD-to-MPS conversion;
5. add streaming transformed-Hamiltonian and MPO construction;

6. cache and checkpoint the symmetry search; and
7. optimize circuit routing and MPS compression.

This sequence preserves the current small-system workflow as a validation case while allowing the reference calculation, entropy analysis, and warm start to be replaced independently as the orbital space grows.